

INFO

Feed

Sat vapor, $F = 170.0 \text{ kmol/h}$

$Z_{\text{ethanol}} = 0.25$ (C2)

$Z_{i\text{-prop}} = 0.25$ (C3) BP = 83°C

$Z_{n\text{-prop}} = 0.30$ (C4) BP = 98°C

$Z_{\text{butanol}} = 0.20$ (C4)

rel vol

w/ n-propanol as reference

$\alpha_{E-np} = 2.17$

$\alpha_{i\text{-p-np}} = 1.86$

$\alpha_{n\text{-p-np}} = 1.00$

$\alpha_{B-np} = 0.412$

constant

1) Ethanol (LNK) 2)

i-propanol (LK)

n-propanol (HK)

n-Butanol (HNK)

5/5

3) component mass bal using feed, comp, dist, bot, mol frac

$$FZ_i = BX_{i,\text{Bot}} + DX_{i,\text{Dist}}$$

$$FZ_{\text{ethanol}} = BX_{\text{ethanol, bot}} + DX_{\text{ethanol, dist}}$$

$$FZ_{i\text{-propanol}} = BX_{i\text{-propanol, bot}} + DX_{i\text{-propanol, dist}}$$

$$FZ_{n\text{-propanol}} = BX_{n\text{-propanol, bot}} + DX_{n\text{-propanol, dist}}$$

$$FZ_{n\text{-butanol}} = BX_{n\text{-butanol, bot}} + DX_{n\text{-butanol, dist}}$$

9/1/00

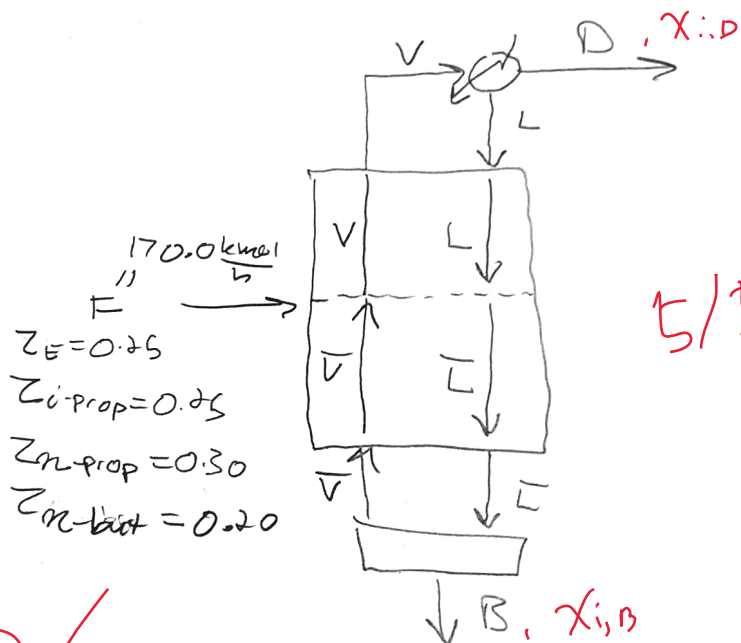
Rayan Baig
CHE313 EXAM 2

Goal

98.0% recovery of i-propanol in the distillate

98.7% recovery of n-propanol in the bottoms

total
has condenser & partial reboiler



5/5

5/5

4) DX_i in distillate, $DX_{i,0.3}$ & 0 value also x_i in dist

Solving for D ✓

assuming LUK $\rightarrow$ 100% to top dist

$$D = \sum (1.0 \times 170.0 \times 0.25) + (0.980 \times 170 \times 0.25) + ((1 - 0.987) \times 170 \times 0.30)$$

$$D = 42.5 + 41.65 + 0.663 = 84.813 \text{ kmol/h}$$

$$D = 84.813 \text{ kmol/h}$$

calculating $\frac{X_i}{\text{mol frac for the components}}$

$$X_i = \frac{DX_i}{D}$$

$$X_{\text{ethanol}} = \frac{FZ_E}{FZ_T} = 1.00 / n_T = 0.501756 \checkmark$$

$$X_{i\text{-prop}} = \frac{(0.980)FZ_{i\text{-prop}}}{FZ_{i\text{-prop}}} = 0.980 / n_T = 0.491721 \checkmark$$

$$X_{n\text{-prop}} = \frac{(1 - 0.987)FZ_{n\text{-prop}}}{FZ_{n\text{-prop}}} = 0.013 / n_T = 0.006523$$

$$X_{n\text{-butanol}} = 0 / n_T = 0$$

$$n_T = 1 + 0.980 + 0.013 = 1.993$$

$$X_{n\text{-prop}} = \frac{0.663}{84.813} = 0.0078$$

sum to 1 ✓

DX_i
↓

$$DX_{\text{ethanol}} = 42.5554 \text{ kmol/h}$$

$$DX_{i\text{-prop}} = 41.7043 \text{ kmol/h}$$

$$DX_{n\text{-prop}} = 0.553221 \text{ kmol/h}$$

$$DX_{n\text{-butanol}} = 0 \text{ kmol/h}$$

$$\sum = 84.813 \text{ kmol/h} \checkmark$$

18/20

$$DX_{i,dist}$$

~~8~~

$$DX_{i,dist} = (Z_i F)(Y-R)$$

$$DX_{ethanol,dist} = (0.25)(170)(1.0) = \underline{42.5 \text{ kmol/h}}$$

$$DX_{i-propyl,dist} = (0.25)(170)(0.987) = \underline{41.65 \text{ kmol/h}}$$

$$DX_{n-prop,dist} = (0.30)(170)(1-0.987) = \underline{0.663 \text{ kmol/h}}$$

$$DX_{n-but,dist} = 0 \text{ kmol/h}$$

$$5) BX_{i,bot}$$

$$BX_{i,bot} = (Z_i F)(Y-R)$$

$$BX_{ethanol,bot} = 0 \text{ kmol/h}$$

$$BX_{i-prop,bot} = (0.25)(170)(1-0.987) = \underline{0.85 \text{ kmol/h}}$$

$$BX_{n-prop,bot} = (0.30)(170)(0.987) = \underline{50.337 \text{ kmol/h}}$$

$$BX_{n-but,bot} = (0.20)(170)(1.00) = \underline{34.00 \text{ kmol/h}}$$

$$\sum BX_{i,bot} = \boxed{B = 85.187 \text{ kmol/h}}$$

$$BX_i \leftarrow \text{mol frac in bot}$$

TEST
B+D=170=F
correct ✓

$$X_{ethanol} = 0 / n_T = \underline{0}$$

$$X_{i-prop} = (1-0.987) / n_T = \underline{0.009965}$$

$$X_{n-prop} = 0.987 / n_T = \underline{0.491779}$$

$$X_{n-but} = 1.00 / n_T = \underline{0.498256}$$

$$n_T = 2.007$$

$$X_{nC3} = \frac{50.337}{85.187} = 0.5909$$

$$X_{nC4} = \frac{34}{85.187} = 0.3991$$

$$BX_i \rightarrow$$

mol frac

$$BX_{ethanol} = 0 \text{ kmol/h}$$

$$BX_{i-prop} = 0.848899 \text{ kmol/h}$$

$$BX_{n-prop} = 41.8932 \text{ kmol/h}$$

$$BX_{n-but} = 42.4449 \text{ kmol/h}$$

6) Find N_{min} & $N_{F,min}$ w/

$LK \rightarrow i\text{-prop}$
 $HK \rightarrow n\text{-prop}$

$$\alpha_{i,prop}^{n\text{-prop}} = 1.86$$

$n\text{-prop}$

$$N_{min} = \ln \left[\frac{(0.980)}{(1-0.980)} \right] / \left[\frac{(11-0.987)}{(10.987)} \right] \bigg/ \ln(1.86)$$

7/10

$$N_{min} = 13.2482 \text{ stages}$$

$$N_{F,min} = \ln \left[\frac{(X_{LK})}{(X_{HK})} \right] \bigg/ \left[\frac{(Z_{LK})}{(Z_{HK})} \right] \bigg/ \ln \alpha_{LK-HK}$$

0.491
 0.007
 ≈ 7.0

$$N_{F,min} = \ln \left[\frac{(0.980)}{(1-0.980)} \right] / \left[\frac{(0.25)}{(0.30)} \right] \bigg/ \ln(1.86) = 6.56509 \text{ stages}$$

$$N_{F,min} = 6.56509 \text{ stages}$$

$$\Delta V_{feed} = F(1-q) \quad q=0$$

$$\Delta V_{feed} = 170 \text{ kmol/h}$$

5/5

8) $\phi = 1.67$ find V_{min} & $(L/D)_{min}$ & D
 $V_{min} = L_{min} + D$ from above in part 4

$$D = 84.813 \text{ kmol/h}$$

$$V_{min} = \frac{(2.17)(42.5)}{2.17-1.67} + \frac{(1.86)(41.65)}{1.86-1.67} + \frac{(1.00)(0.663)}{1-1.67}$$

$$V_{min} = 591.192 \text{ kmol/h}$$

$$L_{min} = 506.379 \text{ kmol/h}$$

$$\left(\frac{L}{D} \right)_{min} = 5.97054$$

15/15

between 1.67 & 2.17

9) $\frac{L}{D} = 1.10 \left(\frac{L}{D} \right)_{\min}$ find N_s & N_F

$$x = \frac{M-1}{\frac{1}{\left(\frac{L}{D}\right)_{\min}} + M} = \frac{1.10-1}{\frac{1}{(5.97054)} + 1.10} = \underline{0.078896}$$

$$\frac{N - N_{\min}}{N+1} = 0.545827 - 0.591422x + \frac{0.002743}{x}$$

plug in x ←

From pt 6

$$N = \underline{29.5712} \text{ stages}$$

$$N_F = N \left(\frac{N_{F,\min}}{N_{\min}} \right)$$

$$N_F = 29.5712 \left(\frac{6.56594}{13.2482} \right)$$

Method is OK!

15/15

$$N_F = \underline{14.6539} \text{ stages}$$

15.63